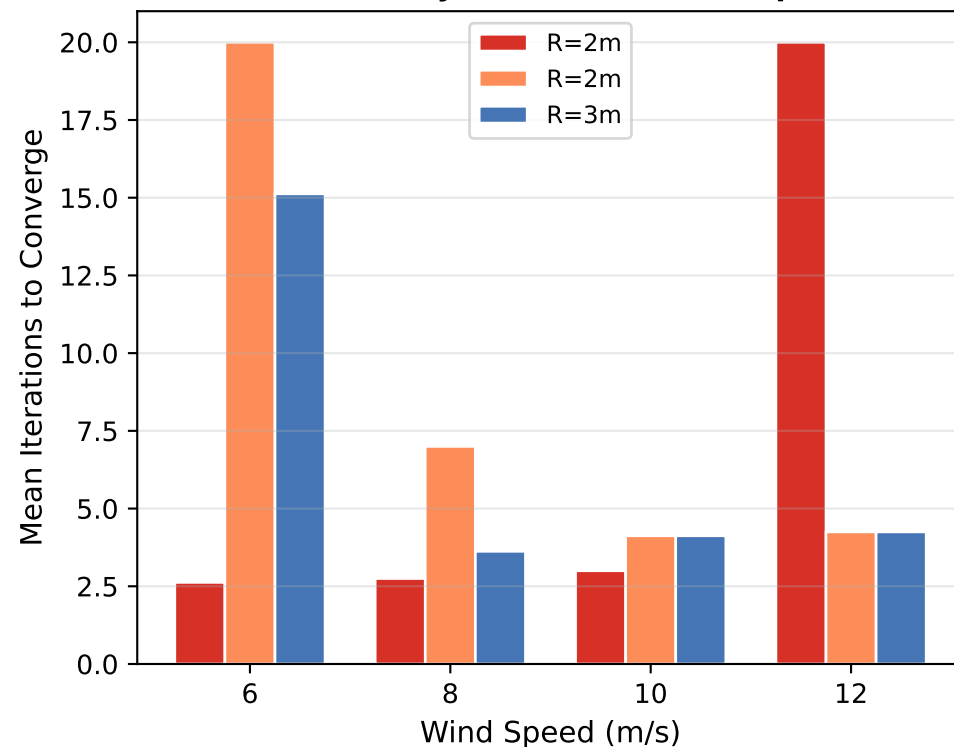
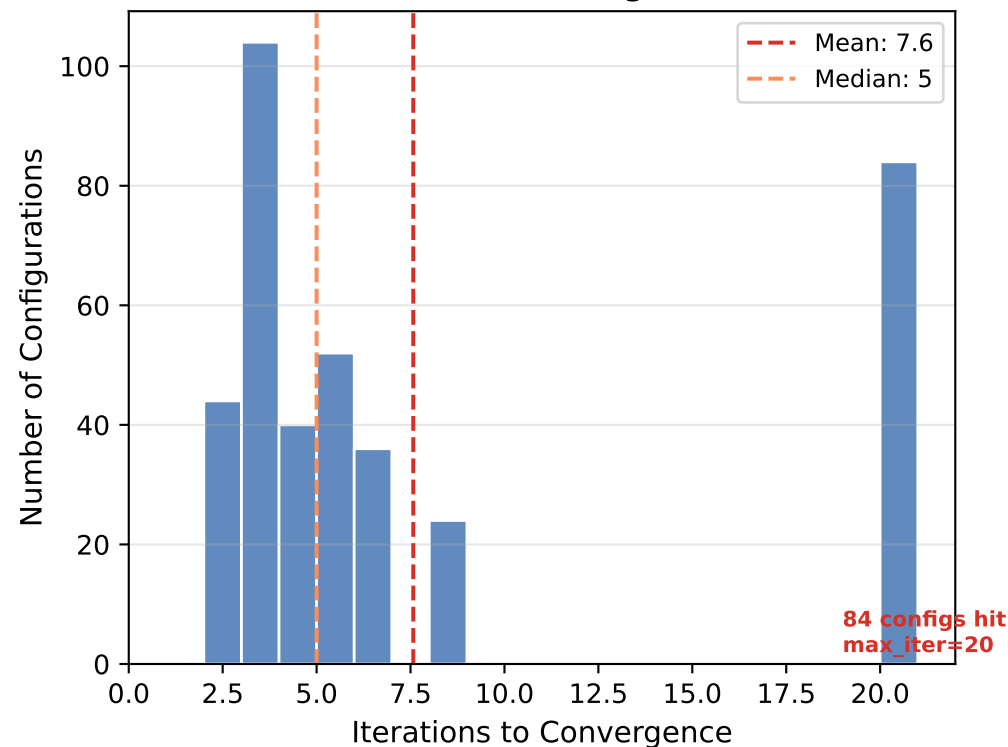


Polygon Solver Convergence: Is It Robust?

BEM v2.1 — Under-Relaxed Jacobi ($\omega=0.5$), $\epsilon=1\times 10^{-6}$)

Solver Iterations Distribution (all 384 configs)

Iterations by Radius and Wind Speed



The polygon solver converges in 8 iterations on average (median 5, range 2–20). R=1.5m configurations converge fastest (2–3 iterations) because tension is zero and angles hit the 5° clamp immediately. R=3.0m at high wind needs more iterations due to stronger thrust-angle coupling. 84 of 384 configurations hit max_iter=20 without full convergence ($\epsilon=0.1^\circ$) — all are R=3.0m at high wind. The 85° angle clamp prevents divergence in all cases. Data: 384-config BEM sweep, under-relaxed Jacobi ($\omega=0.5$).